

DSA LAB 1

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Note: The files submitted in the assignment are filled in by copying my earlier code from my other device where c++ did not work, so the submitted files will have the code copied from the screenshots of the work I did in the online compiler, but no ai is used, all the code is written by me.

Task 1

```
DSALab1Task1.cpp ×
DSALab1Task1 (Global Scope) main()
1 //Lab 1 DSA
2 #include <iostream>
3 using namespace std;
4 int main() {
5     //Task 1 Prediction: All numbers printed properly with a space between them, third will be 7, the rest will be same
6     int numbers[5] = { 2,4,6,8,10 };
7     //Changing third value to 7
8     numbers[2] = 7;
9     //Loop to print as required
10    for (int i = 0; i < 5; i++) {
11        //IF block so no extra space is printed
12        if (i == 4) {
13            cout << numbers[i];
14        }
15        else {
16            cout << numbers[i] << " ";
17        }
18    }
19    return 0;
20 }
```

Output

```
2 4 7 8 10
```

Task 2

Code

```
DSALab1Task2.cpp  What's new
DSALab1Task2
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      //Task 2
6      //Declaration
7      int numbers[5];
8      int total = 0;
9      //Loop for reading numbers into array
10     for (int i = 0; i < 5; i++) {
11         //Process of reading the numbers
12         cout << "Enter a number: ";
13         cin >> numbers[i];
14         cout << endl;
15     }
16     //Loop for adding numbers to total
17     for (int i = 0; i < 5; i++) {
18         total += numbers[i];
19     }
20     //Printing the total
21     cout << "Total = " << total;
22     return 0;
23 }
```

Outputs

```
Microsoft Visual Studio Debug Console
Enter a number: 1
Enter a number: 2
Enter a number: 3
Enter a number: 4
Enter a number: 5
Total = 15
```

```
Microsoft Visual Studio Debug Console
Enter a number: 0
Enter a number: 0
Enter a number: 0
Enter a number: 0
Enter a number: 0
Total = 0
```

```
Microsoft Visual Studio Debug Console
Enter a number: -2
Enter a number: 4
Enter a number: -1
Enter a number: 0
Enter a number: 3
Total = 4
```

The table filled out

Test input	Expected total	Actual total
1 2 3 4 5	Complete before running	15
0 0 0 0 0	Complete before running	0
-2 4 -1 0 3	Complete before running	4

Task 3

Code

```
DSALab1Task3.cpp × DSALab1Task3_.cpp
1  #include <iostream>
2  using namespace std;
3
4  class Student {
5      //Creating public to declare everything under
6  public:
7      //Declaring all needed variables
8      int rollNumber, marks;
9      //Parameterised constructor to initialise the variables
10     Student(int userRollNumber, int userMarks) {
11         rollNumber = userRollNumber;
12         marks = userMarks;
13     }
14     //The display function as required
15     void display() {
16         cout << rollNumber << " " << marks << endl;
17     }
18 };
```

```
18 };
19 int main() {
20     //Task 3
21     //Creating student objects
22     Student s1(1, 80);
23     Student s2(2, 90);
24     cout << "Student 1: ";
25     s1.display();
26     cout << "Student 2: ";
27     s2.display();
28     /*Expected outputs:
29     Student 1: 1 80
30     Student 2: 2 90
31     */
32     return 0;
33 }
```

Output

```
Microsoft Visual Studio Debug Console
Student 1: 1 75
Student 2: 2 90
```

After changing s1.marks to 80

```
Student s1(1,80);
/*Expected outputs:
Student 1: 1 80
Student 2: 2 90
*/
```

Do not expect any change in s2.marks displayed

```
Microsoft Visual Studio Debug Console
Student 1: 1 80
Student 2: 2 90
```

Task 4,5 (Repeated Task)

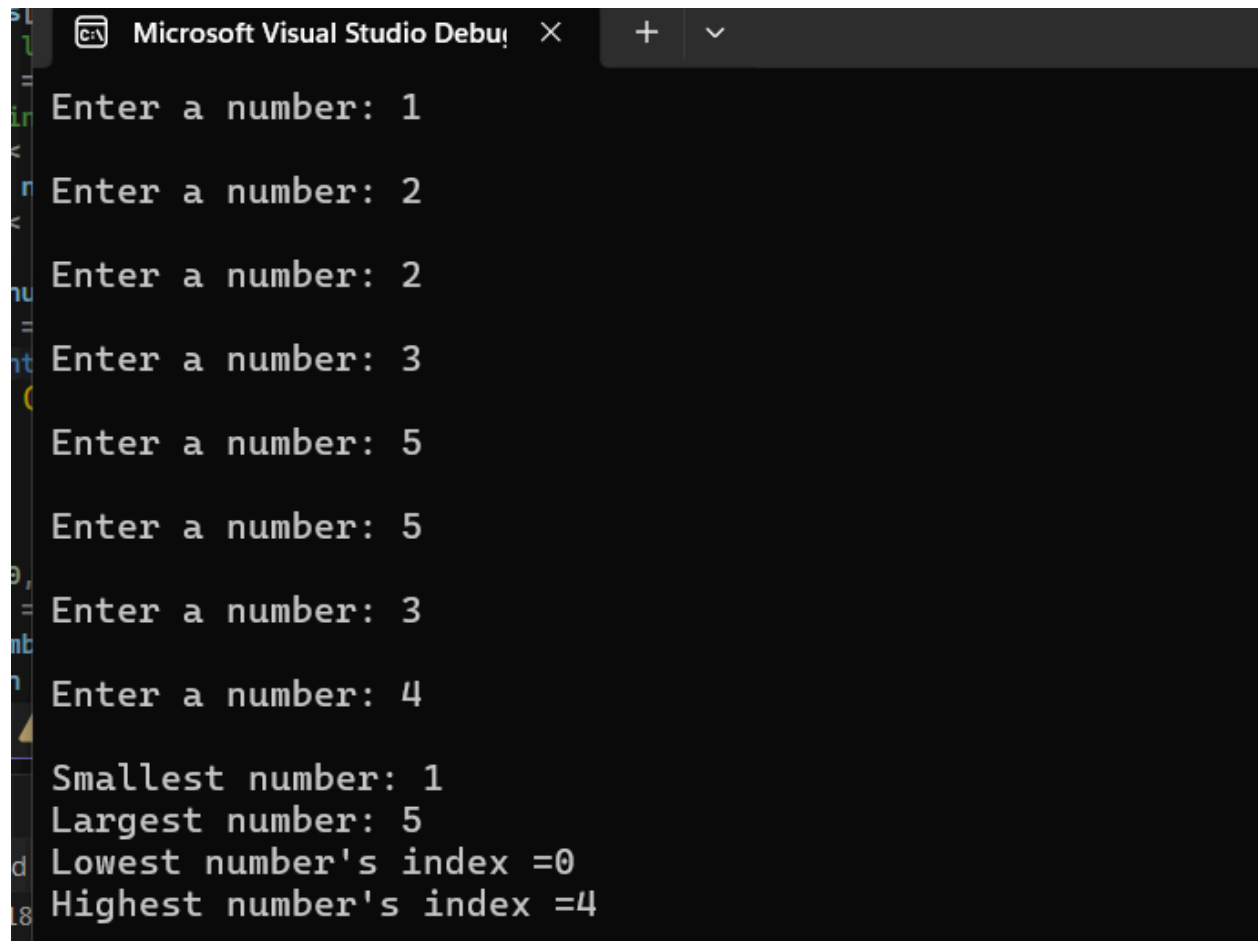
Code

```
DSALab1Task4_5.cpp* X
DSALab1Task4_5 (Global Scope)
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      //Task 4
6      //Declaring array
7      int numbers[8];
8      //Creating loop to read numbers into array
9      for (int i = 0; i < 8; i++) {
10         //Reading numbers into array
11         cout << "Enter a number: ";
12         cin >> numbers[i];
13         cout << endl;
14     }
15     //Declaring array of repeated numbers
16     int repeatnumbers[8];
17     //Using loops to find and store all repeated numbers in array
18     for (int i = 0; i < 8; i++) {
19         for (int c = i + 1; c < 8; c++) {
20             if (numbers[i] == numbers[c]) {
21                 repeatnumbers[i] = numbers[i];
22             }
23         }
24     }
25     //Initialising max as small value and min as large value to use, and variables for index
26     int max = 0, min = 99999, maxIndex, minIndex;
27     for (int i = 0; i < 8; i++) {
28         //Replacing minimum with new minimum if found and replacing minimum value's index
29         if (numbers[i] < min) {
30             min = numbers[i];
31             minIndex = i;
32         }
33         //Replacing maximum with new maximum if found and replacing maximum value's index
34         if (numbers[i] > max) {
35             max = numbers[i];
36             maxIndex = i;
37         }
38     }
39     //Printing the minimum and maximum value
```

```
40 cout << "Smallest number: " << min << endl;
41 cout << "Largest number: " << max << endl;
42 //Initialising the repeat index with out of range index intentionally
43 int minrepeatIndex = 10, maxrepeatIndex = 10, index = 0;
44 //Using do while loop to check if the lowest value has been repeated
45 do {
46     //If lowest value is repeated, it's new index is stored in repeatindex
47     if (repeatnumbers[index] == min) {
48         minrepeatIndex = index;
49     }
50     index++;
51 } while ((repeatnumbers[index] != min) && (index < 8));
52 index = 0;
53 //Same as minimum while loop
54 //Used do while loop to check if the largest value is repeated
55 do {
56     //If repeated, it's new index is stored in repeatindex
57     if (repeatnumbers[index] == max) {
58         maxrepeatIndex = index;
59     }
60     index++;
61 } while ((repeatnumbers[index] != max) && (index < 8));

62 //If repeatindex is changed, print the new index(which is the lowest one)
63 if (maxrepeatIndex != 10) {
64     cout << "Lowest number's index =" << minrepeatIndex << endl;
65 }
66 //If unchanged, print the earlier found index, meaning no repeat
67 else { cout << "Lowest number's index =" << minIndex << endl; }
68 //If repeatindex is changed, print the new index (which is the lowest one)
69 if (maxrepeatIndex != 10) {
70     cout << "Highest number's index =" << maxrepeatIndex << endl;
71 }
72 //If unchanged, print earlier found index, meaning no repeat
73 else { cout << "Highest number's index =" << maxIndex << endl; }
74 return 0;
75 }
```

Outputs

A screenshot of the Microsoft Visual Studio Debug Console window. The window has a dark background and a title bar that says "Microsoft Visual Studio Debug Console". The output text is as follows:

```
Enter a number: 1
Enter a number: 2
Enter a number: 2
Enter a number: 3
Enter a number: 5
Enter a number: 5
Enter a number: 3
Enter a number: 4

Smallest number: 1
Largest number: 5
Lowest number's index =0
Highest number's index =4
```

Non-Repeat version



Microsoft Visual Studio Debug Console



Enter a number: 2

Enter a number: 1

Enter a number: 3

Enter a number: 7

Enter a number: 6

Enter a number: 5

Enter a number: 4

Enter a number: 3

Smallest number: 1

Largest number: 7

Lowest number's index =1

Highest number's index =3

Task 6

Code

```
DSALab1Task6.cpp  ✎  ✕
DSALab1Task6  (Global Scope)

1  #include <iostream>
2
3  using namespace std;
4
5  int main() {
6      //Task 6
7      //Declaring variables and initialising left pointer to 0 index
8      int leftptr = 0, numbers[5], temp = 0;
9      //Loop to read values into array
10     for (int i = 0; i < 5; i++) {
11         //Process of reading values
12         cout << "Enter a number: ";
13         cin >> numbers[i];
14         cout << endl;
15     }
16     //Loop to swap the values either side of middle value
17     while (leftptr != 3) {
18         //Swap
19         temp = numbers[leftptr];
20         numbers[leftptr] = numbers[4 - leftptr];
21         numbers[4 - leftptr] = temp;
22         //Incrementing left pointer
23         leftptr++;
24     }
25     //Loop to display the results
26     for (int i = 0; i < 5; i++) {
27         cout << numbers[i] << " ";
28     }
29     return 0;
30 }
```

Output

```
Microsoft Visual Studio Debug Console
Enter a number: 1
Enter a number: 2
Enter a number: 3
Enter a number: 4
Enter a number: 5
5 4 3 2 1
```

Task 7

Code

```
DSALab1Task7.cpp
DSALab1Task7 (Global Scope)

1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      /*Declaring and initialising variables
6       UniqueCount started from 1 as first value isn't checked
7       and is always considered unique in the design of my code*/
8      int numbers[10], UniqueCount = 1, temp;
9      //Loop to read numbers and fill array
10     for (int i = 0; i < 10; i++) {
11         cout << "Enter number: ";
12         cin >> numbers[i];
13         cout << endl;
14     }
15     //Flag variable for uniqueness
16     bool isUnique = true;
17     //Loop to iterate array
```

```

16     bool isUnique = true;
17     //Loop to iterate array
18     for (int i = 1; i < 10; i++) {
19         //Inner loop checking if it is a repeat
20         for (int c = 0; c < i; c++) {
21             if (numbers[i] == numbers[c]) {
22                 isUnique = false;
23             }
24         }
25         //If not a repeat, swapping with left value
26         if (isUnique == true) {
27             UniqueCount += 1;
28             //Loop to send the unique value to the left most side
29             for (int a = i; a > 0; a--) {
30                 temp = numbers[a];
31                 numbers[a] = numbers[a-1];
32                 numbers[a-1] = temp;
33             }
34             //Results in a horizontally flipped in order unique values
35         }
36         isUnique = true;

```

```

37     }
38     int leftpointer = 0;
39     int rightpointer = UniqueCount - 1;
40     //Using loop to restore original order of unique values
41     while (leftpointer < rightpointer) {
42         //Swapping left with right, meaning swapping the uniques withing themselves
43         temp = numbers[leftpointer];
44         numbers[leftpointer] = numbers[rightpointer];
45         numbers[rightpointer] = temp;
46         leftpointer++;
47         rightpointer--;
48     }
49     for (int i = 0; i < 10; i++) {
50         cout << numbers[i] << " ";
51     }
52     //Printing the count of unique numbers
53     cout << endl;
54     cout << "Unique count: " << UniqueCount;
55     return 0;
56 }

```

```
Microsoft Visual Studio Debug Console
Enter number: 4
Enter number: 2
Enter number: 4
Enter number: 7
Enter number: 2
Enter number: 9
Enter number: 7
Enter number: 1
Enter number: 9
Enter number: 1
4 2 7 9 1 4 2 7 9 1
Unique count: 5
```

Task 8

Note: I have completed all the tasks as separate projects in visual studio, so according to my understanding of the task, I have placed just the cpp files in the folder and will commit them individually.

```
C:\Users\rjhas>git config --global user.name "MuhammadShayan2590"
```

```
C:\Users\rjhas>cd source  
C:\Users\rjhas\source>cd repos  
C:\Users\rjhas\source\repos>cd DSALab1  
C:\Users\rjhas\source\repos\DSALab1>|
```

```
C:\Users\rjhas\source\repos\DSALab1>git init  
Initialized empty Git repository in C:/Users/rjhas/source/repos/DSALab1/.git/  
  
C:\Users\rjhas\source\repos\DSALab1>git status  
On branch master  
  
No commits yet  
  
Untracked files:  
  (use "git add <file>..." to include in what will be committed)  
    DSALab1Task2.cpp  
    DSALab1Task3.cpp  
    DSALab1Task4_5.cpp  
    README.md  
  
nothing added to commit but untracked files present (use "git add" to track)
```

```
C:\Users\rjhas\source\repos\DSALab1>git add DSALab1Task2.cpp  
  
C:\Users\rjhas\source\repos\DSALab1>git add DSALab1Task3.cpp  
warning: in the working copy of 'DSALab1Task3.cpp', LF will be replaced by CRLF the next time Git touches it  
  
C:\Users\rjhas\source\repos\DSALab1>git add DSALab1Task4_5.cpp  
warning: in the working copy of 'DSALab1Task4_5.cpp', LF will be replaced by CRLF the next time Git touches it  
  
C:\Users\rjhas\source\repos\DSALab1>git status  
On branch master  
  
No commits yet  
  
Changes to be committed:  
  (use "git rm --cached <file>..." to unstage)  
    new file:   DSALab1Task2.cpp  
    new file:   DSALab1Task3.cpp  
    new file:   DSALab1Task4_5.cpp  
  
Untracked files:  
  (use "git add <file>..." to include in what will be committed)  
    README.md
```

```
C:\Users\rjhas\source\repos\DSALab1>git add README.md
```

```
C:\Users\rjhas\source\repos\DSALab1>git status
```

```
On branch master
```

```
No commits yet
```

```
Changes to be committed:
```

```
(use "git rm --cached <file>..." to unstage)
```

```
new file:   DSALab1Task2.cpp
```

```
new file:   DSALab1Task3.cpp
```

```
new file:   DSALab1Task4_5.cpp
```

```
new file:   README.md
```

```
C:\Users\rjhas\source\repos\DSALab1>git commit -m"Documents and test results"
```

```
[master (root-commit) 950d869] Documents and test results
```

```
4 files changed, 151 insertions(+)
```

```
create mode 100644 DSALab1Task2.cpp
```

```
create mode 100644 DSALab1Task3.cpp
```

```
create mode 100644 DSALab1Task4_5.cpp
```

```
create mode 100644 README.md
```

```
C:\Users\rjhas\source\repos\DSALab1>git log --oneline
```

```
950d869 (HEAD -> master) Documents and test results
```